



ZERO-WASTE KITCHEN

& Pantry Expiry Assistant

"Know what you have. Use it before it's gone."

Unit	ITS203 — Object-Oriented Design and Programming
Assessment	Assessment C — Milestone 1: Project Idea Proposal
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1 • Problem Description

Honestly, I believe food wasting is something that most homes suffer with. We've all purchased goods with good intentions, only to discover a bag of spinach or a jar of sauce lurking way in the back of the fridge far beyond its expiration date. It's so annoying because it simply seems like throwing money away.

This software is for normal individuals, home cooks or busy students wanting to keep a closer watch on what they really have in their kitchen. It makes it easier to keep track of inventory and expiration dates, helping individuals use up what they have before it spoils, saving money and reducing wasteful waste.

2 • Proposed Idea

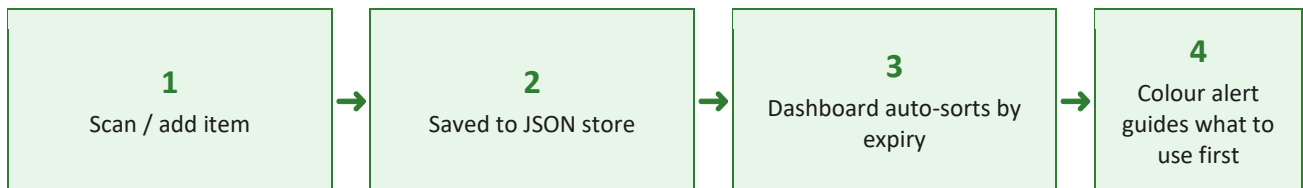
To achieve this, I am developing a C# Windows Forms desktop application called the Zero-Waste Kitchen & Pantry Expiry Assistant.

Basically, it's a digital pantry. You may also record groceries into the app when you get them home, filling in categories, quantities, and expiry dates. The primary dashboard will list everything you own, auto-sorted so the things closest to expiry jump up right at the top. I want to have colour coding, like red for expired products, yellow for things that are going to go off soon and green for fresh stuff, so it's evident what you need to prepare first. You may also quickly adjust amounts or remove things when they are used up.

A simple traffic-light colour system keeps it glanceable:

Status	What it means on the dashboard
 Expired	Past its expiry date — flagged first, top of the list.
 Expiring soon	Within a configurable warning window (e.g. 3 days) — use it next.
 Fresh	Comfortably within date — no action needed yet.

How it flows



3 · Inspiration

I choose this project since it's a true issue I have in my own kitchen all the time. I've used phone notes and regular spreadsheets previously, but they're clumsy and easy to forget. Building a custom Windows Forms app seemed like a great fun and useful method to solve a daily aggravation and to put the object-oriented programming ideas we are studying in class to work.

4 · Functional Requirements

- + Add new items — Users may add food products by entering a name, choosing a category (Dairy, Produce, Pantry) specifying a quantity and inputting an expiry date.
-  Live inventory dashboard — A graphical representation of data grids, listing all items currently being tracked in real-time.
-  Expiry alerts & colour coding — the programme will calculate the days remaining for an item to expire and will highlight it visually so that critical items are brought to front.
-  Update & delete — Users may easily update the quantity of an item at any time (e.g. if they drank half a carton of milk) or remove it altogether when it has been used up.
-  Category filtering — A simple filter drop down menu that enables users to look at just the goods in the refrigerator, only the things in the pantry or certain species of food.

5 · Non-Functional Requirements

-  Performance — While surfing between tabs or filtering listings, there should be no delay in the programme. The performance of the app should be such that it seems speedy.
-  Usability — The user interface must to be simple and easy to understand. It needs to be something that each and every person living in a house can open and figure out how to add or check an item without the need for a handbook.
-  Dependability — It should not crash if someone accidentally inputs an odd date or leaves a text field empty; it should have adequate error handling to gracefully manage such failures. This is a need for reliability.

6 • Data Storage Plan

I am intending on using local JSON file storage to save the data instead of creating a complete database server.

When the app is opened, it will load the stored items from a local JSON file into the program's memory. The user closes the programme (or presses a save button) and it will serialise the current list and update the file. Since it is a single project for a brief trimester, the file-based JSON method makes everything lightweight, simple to test and fully self-contained on the machine.